Spectral–Spatial Depth-Based Framework for Hyperspectral Underwater Target Detection

Qi Li, Jinghua Li, Tong Li, Zheyong Li, and Pei Zhang

***Abstract*— Ocean-related research is of critical significance to the national marine military force. Hyperspectral underwater target detection (HUTD) has attracted widespread attention in recent years. However, most of the previous methods only relied on the spectral features of underwater targets and did not fully exploit their spatial characteristics. To address the issue, a spectral–spatial depth-based framework is proposed, which utilizes the 3-D convolution operation to capture spectral–spatial features and gains finer detection based on predicted depth. In particular, the proposed framework adopts the data transferring network to remove the noise interference by transferring the real hyperspectral data into corresponding synthetic data, which are exploited to train models. Then, considering that underwater target spectra highly depend on its depth in water, the depth estimation network is utilized to predict an accurate depth of a target, which can contribute to selecting a suitable detection network and gaining a general contour of the target. Since the underwater target spectrum is jointly determined by the target and the surrounding water column, the spectral–spatial detection network extracts the spectral–spatial features for underwater target detection. Using a pool dataset, sea dataset, and synthetic hyperspectral image (HSI), we evaluate the performance of the proposed framework in terms of receiver operating characteristic (ROC) curve and area under the ROC curve (AUC) value, both qualitatively and quantitatively. Meanwhile, extensive detection experiments demonstrate the robustness and effectiveness of the TDSS-UTD over several state-of-the-art methods.**

***Index Terms*— Deep learning, hyperspectral image (HSI), spectral–spatial features, synthetic dataset, underwater target detection (UTD).**

I. Introduction

MARINE military forces have been a significant portion of the national armed forces, and their extensive deployment has encouraged ocean-related research, of which study on underwater detection is essential [1], [2], [3], [4], [5], [6], [7], [8]. Hyperspectral imagery is frequently employed in agriculture [9], [10], [11], geoscience [12], and environmental monitoring [13], [14]. In complex marine environments, hyperspectral image (HSI), which contains more information than standard RGB imagery, are better suited for underwater target detection (UTD) [15], [16], leading to widespread interest in hyperspectral UTD (HUTD).

Manuscript received 10 January 2023; revised 20 February 2023 and 6 April 2023; accepted 6 May 2023. Date of publication 11 May 2023; date of current version 23 May 2023. *(Corresponding author: Jinghua Li.)*

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Digital Object Identifier 10.1109/TGRS.2023.3275147

The main purpose of HUTD is to overcome the interference of the water environment on target characteristics and detect interest underwater targets. Compared to land-based hyperspectral target detection (HTD) methods, little existing research has been done for HUTD. Due to the difference between the environment of the underwater environment and above the water surface, the propagation of light can be affected by water absorption and scattering, leading to the characteristic of the HSI for underwater targets different from that of land-based targets. Moreover, the spectra of the underwater target to be detected are highly dependent on the inherent optical properties of the water and the depth of the target [17]. Thus, it is not reasonable to apply land-based HTD methods directly to the UTD task. Therefore, it is necessary to study HUTD.

In the field of HTD, many representative traditional algorithms have been proposed [18], [19]. Most early traditional algorithms [20], [21], [22], [23], including reed-xiaoli (RX), matched filter (MF), spectral angel mapping (SAM), orthogonal subspace projection (OSP), and constrained energy minimization (CEM), employ some spectral characteristics of the target or background as a priori knowledge to detect targets. RX [24], [25] separates the target from the background and noise by statistically evaluating the anomalies, whereas MF [26] builds a matching filter to distinguish the background and the target using the priori information of target and background. Using the priori target information, the SAM [27] finds the spectral angular distance between the pixels to be tested and the known target. To reduce interference from the background, OSP [28], [29] creates a subspace and projects the test pixel to the orthogonal subspace of the background spectrum. It then selects the vector with the highest signal-to-noise ratio (SNR) among the remaining target features. CEM [30] exploits a finite impulse response filter to achieve the separation of the target and the background. In addition, traditional machine learning techniques, such as support vector machine (SVM) [31], [32] and random forest (RF), are also exploited for HTD.

Deep learning has acquired prominence in many disciplines over the past decade, and it performs incredibly well in HTD [33], [34]. Convolutional neural network (CNN) is the most common class of network models, and numerous features can be automatically extracted by the hidden layer. The deep convolutional neural network (DCNN) model [35], [36] was the first application of CNN to HTD. The DCNN feeds the 1-D spectral data into the network for training, extracts spectrum characteristics, and then creates a classification

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network. By adding a reshape layer, which enables the network to extract spectral and spatial features to significantly increase classification accuracy, HSI-CNN [37] enhances the CNN. Some studies use sophisticated neural networks to extract features; however, these techniques are usually ineffective due to the limited training dataset and the imbalanced distribution of targets and backgrounds.

The application of transfer learning in the field of HTD is a relatively active research area. Transfer learning is a machine learning method that allows the use of knowledge obtained in one task to be used in another. In the field of HTD, transfer learning can help improve the accuracy of target detection by training deep learning models. Cheng and Malhi [38] developed transferring learning for classification. The experimental results show that this method can effectively improve the accuracy of hyperspectral classification accuracy. Nyabuga et al. [39] employed transfer learning to train a CNN to classify targets in HSIs.

All the aforementioned techniques are based on land-based HTD; however, the underwater observed spectrum is determined by the surrounding water environment and the target with its depth. The land-based radiative transfer equation (RTE) and underwater RTE [17] are distinctive. It is not feasible to immediately utilize land-based HTD techniques in the underwater environment [40], [41], [42], [43], [44]. Due to the high confidentiality and the shortage of existing studies about UTD, there are limited references accessible. Gills [17] constructed a target space of all possible observed targets and then separated the target from the background by projecting the combination of image data and the target space into a low-dimensional space. Without considering the physical model used in hyperspectral remote sensing of the ocean, Fu et al. [45] directly applied the CEM to HUTD. Qi et al. [46] applied the bathymetric model to band selection, while target detection remained with the CEM.

Even while some techniques have achieved excellent performance in HUTD [47], [48], they still face several limitations. First, only a few HUTD methods take the underwater RTE into account, and the majority of them directly use land-based detection methods for UTD. As a result, these models have low applicability underwater, and their performance is highly susceptible to the influence of water. Second, a spectral–spatial feature extraction model must be developed because the target and the nearby water body both contribute to the target spectrum.

In this article, we address the above limitations and propose the TDSS-UTD for HUTD. It primarily comprises the spectral–spatial detection network, the depth estimation network, and the data transferring network. To construct the synthetic dataset and synthetic HSI required for our training model, at the very beginning, we input the optical deep water data into IOPE-Net [43] to obtain the properties of the water. The noise interference brought by other pollutants, such as water particles, is then removed by the data transferring network, which transfers the real hyperspectral data into the corresponding synthetic data. The depth value of the underwater target is predicted using the trained depth estimation network, which facilitates our model to use a more suitable spectral–spatial detection network. Finally, we divide the input HSI into several cubic data, detect the cubic data using a spectral–spatial detection network, and derive the predicted class probability map corresponding to the HSI.

In summary, the main contributions of the proposed method in this article can be listed as follows.

1) Due to the lack of hyperspectral underwater target datasets, we construct the synthetic dataset to provide support for the training of network models. Then, we develop the data transferring network to reduce the interference of the water environment in the HUTD task.
2) The synthetic dataset is used to train the depth estimation network, which is devoted to selecting a finer detection network and obtaining general underwater target contour information. To the best of our knowledge, this is the first time that hyperspectral techniques are applied to obtain the depth and contour of underwater targets.
3) Considering the special dependency of the observed spectrum of the underwater target, the systematic combination of the 3-D convolution layer and 2-D convolution layer is proposed in the spectral–spatial detection network for extracting spectral–spatial information in HSI, thereby significantly improving the detection accuracy. Experiments on three datasets verify the superiority of the proposed framework.

The remainder of this article is organized as follows. Section II briefly introduces some related work. In Section III, we elaborate on the proposed method in this article. In Section IV, the experimental results and parameter sensitivity of the model are presented, and the robustness of the model is confirmed. Finally, the conclusions of this article are provided in Section V.

II. Related Work

A. Underwater RTE

The airborne hyperspectral camera system for UTD generally contains three parts: light source, underwater target, and airborne hyperspectral imaging system. Since the normal natural light source will suffer a more serious loss in water transmission, in this experiment, we use the natural light source with good lighting conditions. The underwater target is the object we are interested in, which is to be detected. The underwater HSI captured by the hyperspectral sensor on the aircraft is transmitted through air and water column, and the reflected spectral information of any target in a certain area is collected. We disregard the reflectance loss in the air due to the spectral imager being vertically overhead at a height of 3.3 m above the water surface during our experiments; hence, no atmospheric correction is applied, and the solar incidence angle is set to zero. The reflectance is adjusted by the calibrated whiteboard, which is demonstrated in Section IV.

According to the literature [17], the reflectance for each pixel in an underwater HSI can be described as

$$r(\lambda) = r_\infty(\lambda)\left(1 - e^{-\left(k_d(\lambda)+k_u^c(\lambda)\right)H}\right) + r_B(\lambda)e^{-\left(k_d(\lambda)+k_u^b(\lambda)\right)H} \quad (1)$$

where $r(\lambda)$ denotes the value of reflectance at wavelength λ, $r_\infty(\lambda)$ and $r_B(\lambda)$ refer to the reflectance of pure water and

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substances in water, respectively, and H indicates the value of the depth at which the underwater target is. Also, $k_d(\lambda)$, $k_u^c(\lambda)$, and $k_u^b(\lambda)$ are the attenuation coefficients in water, which can be expressed nonlinearly by the water properties absorption $a(\lambda)$ and scattering $b_b(\lambda)$ as follows:

$$\begin{aligned} u &= b_b(\lambda)/(a(\lambda)+b_b(\lambda)) \\ k &= a(\lambda)+b_b(\lambda) \\ r_\infty(\lambda) &\approx (0.084+0.170\ u)u \\ k_u^c(\lambda) &\approx 1.03(1+2.4\ u)^{0.5}k \\ k_u^b(\lambda) &\approx 1.04(1+5.4\ u)^{0.5}k \\ k_d(\lambda) &= (a(\lambda)+b_b(\lambda))/\cos\theta \end{aligned} \tag{2}$$

where k denotes the sum of the absorption and scattering rate of the water column, called the absorption coefficient. It indicates the degree of attenuation of light in the water column, that is, the optical resistance encountered when the light propagates in the water column. The larger the absorption coefficient, the light in the body of water, the faster the attenuation.

The aforementioned model illustrates that the observed spectra depend not only on the reflectance spectrum of the target and its depth value but also on the water conditions around the target. We can determine the real spectrum of the target if we know the depth at which it is located, its reflection spectrum, and the water quality characteristics of the absorption rate and scattering rate.

B. Synthetic Dataset

We construct a theoretical hyperspectral dataset of underwater targets (named the synthetic dataset in this article) from the underwater RTE. Also, we use the synthetic dataset to train our model, due to the fact that there is no publicly available hyperspectral dataset on underwater targets and the number of our real experimental data is far from the amount of data required to train a deep learning network.

Detection of underwater iron substances has a wide range of implications, including finding and clearing submarine obstacles, detecting and assessing pollution, finding and exploiting submarine resources, and preserving artifacts and historical sites. Also, our research work is originally intended for underwater vehicle detection. The research on the detection of underwater iron substances is of representative significance. Therefore, the focus of this study is on the development of hyperspectral-based techniques for the detection of underwater iron substances. It is first necessary to estimate the water properties absorption $a(\lambda)$ and scattering $b_b(\lambda)$ by inputting the optical deep water data into the IOPE-Net [43].

After that, we collect the iron spectral data and provide a series of depth values H of iron in the water (0.1 m for the range of 0–1.0 m and 0.3 m for the range of 1.0–1.9 m). We then construct the synthetic dataset by the underwater RTE. As shown in Fig. 1, it is obvious that our synthetic dataset is not disturbed by factors such as underwater noise, considering the construction method.

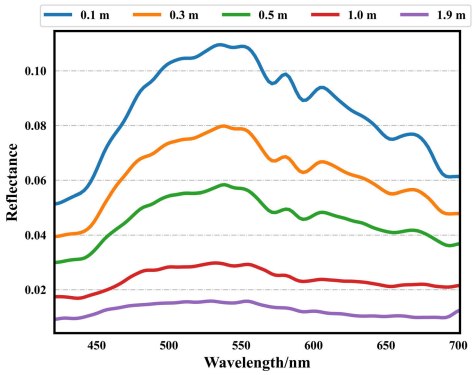


Fig. 1. Synthetic underwater target spectra at various depths. The energy is absorbed more as the light travels deeper into the water. Therefore, the reflectance value decreases with the depth value rises.

C. Hyperspectral Data Preprocessing

1) Derivative Spectra: The amplitude of the spectral curve essentially declines after the depth is greater than 1.6 m, leading to little variation in amplitude between the spectral curves. The derivative spectra could characterize some minor changes in the spectral contours and act as "amplifying characteristics" in comparison to the original spectra. As a result, in this study, we additionally characterize the trend of the spectral curves using the second-order derivative spectra and evaluate the similarity of the spectral curves by combining the tendency represented by the derivative spectra and the amplitude difference of the original spectral curves

$$r'(\lambda_i) = \frac{r(\lambda_{i+1}) - r(\lambda_{i-1})}{2\Delta\lambda} \tag{3}$$

$$\widetilde{r}(\lambda_i) = \frac{r'(\lambda_{i+1}) - r'(\lambda_{i-1})}{2\Delta\lambda} \tag{4}$$

where $r(\lambda_i)$ represents the reflectance at wavelength λ_i, $r'(\lambda_i)$ and $\widetilde{r}(\lambda_i)$ refer to the first-order derivative spectral value and the second-order derivative spectral value, respectively, and $\Delta\lambda$ denotes the wavelength interval.

2) Cubic Data: HSI in Fig. 2(a) is 3-D data that x and y characterize the spatial geolocation and λ represents the spectral dimension. To efficiently use the spatial and spectral information in the hyperspectral data, we segment a series of cubic data from HSI and fed the cubic data into the spectral–spatial detection network in our proposed framework to extract the spectral–spatial characteristic, which could then be applied to detection.

The number of cubic data segmented from an HSI is

$$\text{Num} = (N - N_n + 1) \times (M - N_n + 1) \tag{5}$$

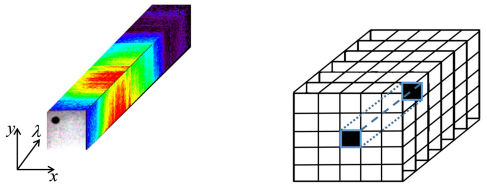
where N and M represent the height and width of the HSI, respectively, and N_n denotes the neighborhood size.

In this experiment, the neighborhood sizes of the cubes were set to 5, 7, 9, 11, 13, and 15. Each set of cubic data labels is determined by the most center pixel, and Fig. 2(b) shows the cube data with a neighborhood size of 5 as an example.

III. Proposed Method

In this study, our proposed TDSS-UTD framework mainly consists of the data transferring network, the depth estimation

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(a) (b)

Fig. 2. (a) Hyperspectral underwater image. (b) Cubic data split from the Hyperspectral underwater image with a neighborhood size of 5. The label of the cubic data is determined by the most central pixel.

network, and the spectral–spatial detection network. The output of the IOPE-Net serves as the basis for the TDSS-UTD framework, and water properties estimated by the IOPE-Net are utilized to create a synthetic dataset, which is used to train our proposed model. The workflow of the TDSS-UTD framework for HUTD is shown in Fig. 3. The data transferring network first transfers the real hyperspectral data into the corresponding synthetic data, which is exploited as the input of the following network model. This removes external interference such as electronic noise or particle scattering making our model more robust. Moreover, the depth estimation network is trained using the data and labels of the synthetic dataset, where the sum of the mean square error (mse) of the depth, the retrieved spectra, and its derivative spectra is used as the loss function. Using the trained depth estimation network, we can determine the depth value of the underwater target. Finally, we segment the input HSI into several cubic data, where the label of cubic data is determined by its central pixel. These cube data are fed into the spectral feature extraction subnetwork and the spatial feature extraction subnetwork to obtain the spectral–spatial features. We then reshape the spectral–spatial features into a 2-D matrix, use the CNN to predict the category of the cube data, and then obtain the detection map of the whole HSI.

A. Data Transferring Network

The underwater HSIs collected during our experiments are frequently noisy due to the interference of particle reflection and scattering in water. These noises are distributed over a wide range of wavelength bands, which will seriously impair the performance of the analytical model. Therefore, we design a data transferring network to transfer the noisy real data to noise-free synthetic data.

The realistic and synthetic data at depths of 0.1, 0.8, 1.6, and 1.9 m are compared in Fig. 4. In the [400, 500] nm band, the amplitude differences between the real and synthetic data at depths of $[0, 0.8]$ m are massive. However, the amplitude differences and curve waveform discrepancies between the real and synthetic data at depths of [0.8, 1.6] m are both minimal. Moreover, the amplitudes of the real and synthetic data are both extremely small at depths of [1.6, 1.9] m.

These differences between the real spectra and the synthetic spectra are caused by the noise interference in the instrumentation and underwater optical transmission process. The synthetic spectra represent the theoretical reflectance value that iron should exhibit in the current water environment and depth. The real spectra are obtained by the sensor during the experiment. Consequently, we set up a data transferring network for three depth intervals, $[0, 0.8]$, $[0.8, 1.6]$, and $[0.8, 1.6]$ m, by examining the disparity between the synthetic data and the real data.

Transfer learning typically uses feature extraction to transfer features. Feature extraction is a machine learning technique that aims to extract useful characteristics from the original data and use these characteristics for another model. In transfer learning, features are extracted by neural networks pretrained on a source task, and the features are utilized to target task. In our study, this source task is to train the data transferring network with known noisy real data and its corresponding noise-free synthetic data. Then, the trained data transferring network is used to extract features for transferring the novel and noisy real data to noise-free synthetic data. The network structure of the data transferring network we developed, which is a supervised regression network, has a total of three fully connected layers. For an input real spectral curve, the feature extracted from the data transferring network would transform it into a predicted spectral curve

$$\begin{aligned} y^{(1)} &= f\left(w^{(1)} * x + b^{(1)}\right) \\ y^{(i)} &= f\left(w^{(i)} * y^{(i)} + b^{(i)}\right), \quad i = 2, 3 \\ \hat{y} &= S\left(y^{(3)}\right) \end{aligned} \tag{6}$$

where $f(\bullet)$ denotes the nonlinear operation, $w^{(i)}$ denotes the weight of the ith layer, $b^{(i)}$ denotes the bias term of the ith layer, x denotes the true spectral curve of the input, $y^{(i)}$ denotes the output of the ith layer in the network, S denotes the sigmoid operation, and $\hat{y}$ is the predicted spectral curve. In this article, a hybrid loss function is defined for the data transferring network with three components: the spectral angular distance of the synthetic and predicted spectral curves, the mse of the synthetic and predicted spectral curves, and the mse of the corresponding spectral derivatives. This hybrid loss function is defined as

$$L_T = L_m(y, \hat{y}) + \lambda_{1t} L_s(y, \hat{y}) + \lambda_{2t} L_d(\widetilde{y}, \widetilde{\hat{y}}) \tag{7}$$

where y and $\hat{y}$ refer to the synthetic spectral curve corresponding to the true spectral curve and the predicted spectral curve, respectively, λ_{1t} and λ_{2t} are the weights of the hybrid loss function; L_m denotes the loss function to constrain the curve magnitude difference, and L_s and L_d represent the loss function to capture the curve trend difference.

The curve magnitude differences are constrained using the mse function as follows:

$$L_m = \frac{1}{N} \sum_{i=1}^{N} (y_i - \hat{y}_i)^2 \tag{8}$$

where N denotes the number of total bands and y_i and $\hat{y}_i$ refer to the value on the ith band for the true spectral curve and the predicted spectral curve, respectively.

The curve trend differences are constrained using the spectral angle and the mse of the corresponding derivative spectrum

$$L_d = \frac{1}{N} \sum_{i=1}^{N} (\widetilde{y}_i - \widetilde{\hat{y}}_i)^2 \tag{9}$$

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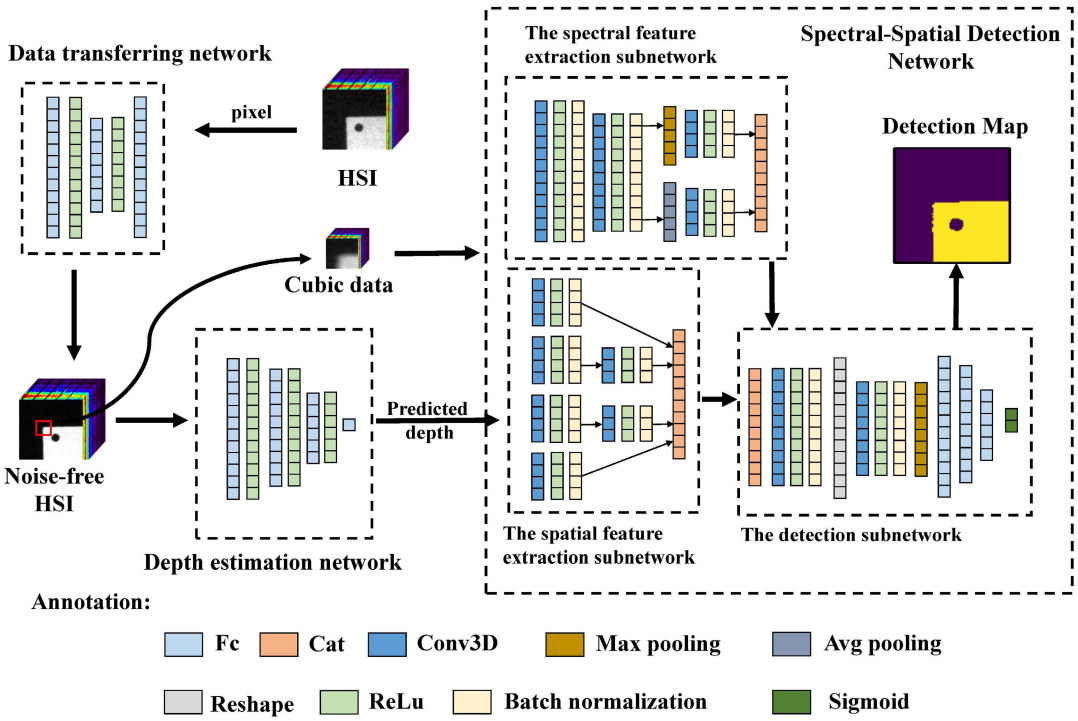


Fig. 3. Workflow of the TDSS-UTD framework for HUTD. First, the real HSI is fed into the data transferring network to remove external interference. After that, the depth estimation network predicts the depth of the underwater target in the noise-free HSI. Finally, we can get the detection map from the spectral–spatial detection network.

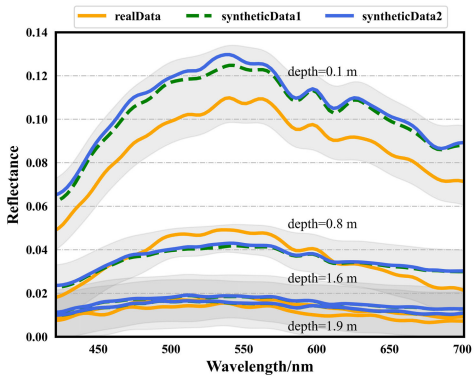


Fig. 4. Comparison of real data and synthetic data at depths of 0.1, 0.8, 1.6, and 1.9 m. The noise interference causes the difference between the real spectra and the synthetic spectra.

$$L_s = \arccos\left(\frac{\langle y, \hat{y}\rangle}{\|y\|, \|\hat{y}\|}\right) \tag{10}$$

where $\widetilde{y}_i$ and $\widetilde{\hat{y}}_i$ denote the value on the ith band for the second-order derivative spectra of the synthetic spectral curve and the predicted spectral, respectively, which characterize the trend of the curves.

The stochastic gradient descent (SGD) optimization algorithm is used to continuously update the weights and minimize the loss function so that the output of the data transferring network converges to the corresponding synthetic data.

B. Depth Estimation Network

To divide a finer detection network and gain a general contour of the target, it is necessary to know the depth information where the target is located during the detection process. We developed the depth estimation network, which has essentially the same network structure as the data transferring network, where only the output is changed to a node. The network's output is $\hat{h} = f(r)$.

Once more, the mse of the actual and predicted depths, the mse of the input spectra and the retrieved spectra, and the mse of the corresponding derivative spectrum are all components of the hybrid loss function of the network

$$L_D = L_m(h, \hat{h}) + \lambda_{1d} L_m(r, \hat{r}) + \lambda_{2d} L_m(\widetilde{r}, \widetilde{\hat{r}}) \tag{11}$$

where h represents the actual depth of the synthetic data, $\hat{h}$ is the predicted depth of the synthetic spectral curve r, and r and $\hat{r}$ denote the input synthetic spectral curve and the retrieved spectral curve with the predicted depth $\hat{h}$, respectively. The corresponding second-order derivative spectra are $\widetilde{r}$ and $\widetilde{\hat{r}}$. The weights of the depth estimation network model are also updated by employing the SGD optimization algorithm as follows:

$$\theta \leftarrow \theta + l\frac{\partial L_D}{\partial \theta}. \tag{12}$$

C. Spatial–Spectral Detection Network

To effectively utilize the accurate depth information predicted by the depth estimation network, we construct a spectral–spatial detection network at each depth (0.1–1.0, 1.3, 1.6, 1.9) m so that we can further improve the detection accuracy.

As previously mentioned, the spectral–spatial detection network proposed in this research has three subnetworks: a subnetwork for extracting spatial features, a subnetwork for

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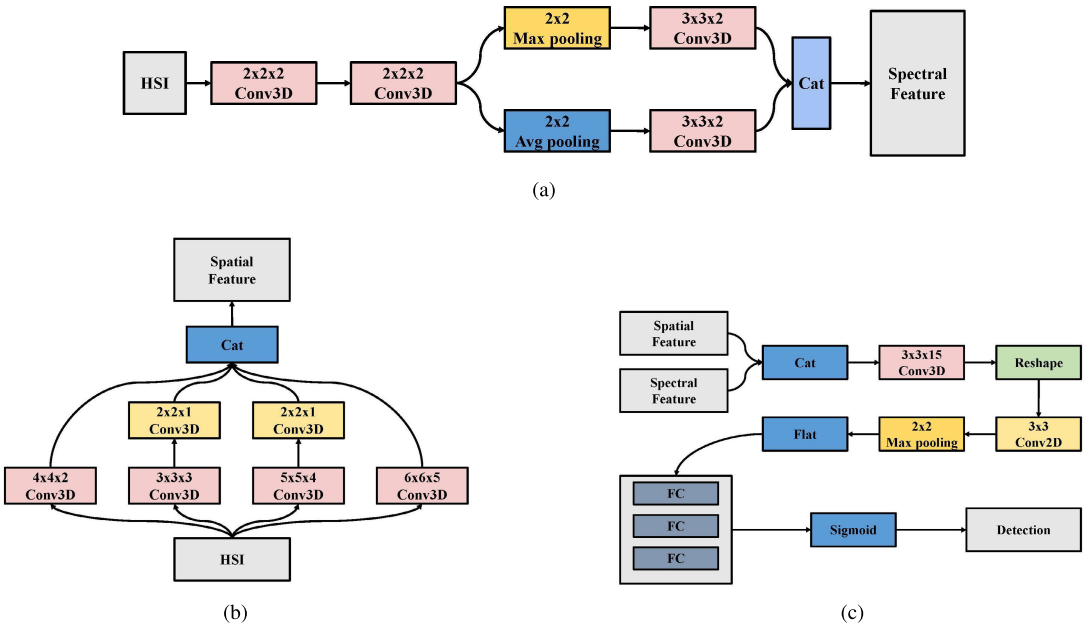


Fig. 5. Structure diagram. (a) Spectral feature extraction subnetwork. (b) Spatial feature extraction subnetwork. (c) Detection subnetwork.

extracting spectral features, and a subnetwork for extracting detection characteristics. We use special structures to extract special features. For instance, we utilize a 3-D convolution operation to extract spectral features and a pooling operation to extract spatial information.

Cubic data are input into the spectral–spatial detection network. First, the two types of features obtained from the spatial feature extraction subnetwork and the spectral feature extraction subnetwork are concatenated together. The feature matrix on each channel is output into a 1-D array using the 3-D convolution operation. After that, multiple 1-D array features are combined into an image-like 2-D matrix using the reshape operation, which can fully use the extracted spectral–spatial feature. Finally, the detection map is determined by the detection subnetwork.

The model topology of the spectral feature extraction subnetwork is shown in Fig. 5(a). To extract the deep spectral feature, the input cubic data are processed by two 3-D convolutional layers with a kernel size of $2 \times 2 \times 2$. After that, there are a maximum pooling layer and an average pooling layer to decrease the feature scale. Finally, the characteristic matrix is extracted by one 3-D convolutional layer with a kernel size of $3 \times 3 \times 2$, which is then concatenated into spectral features.

Fig. 5(b) shows the structure diagram of the spatial feature extraction subnetwork. The input cubic data are extracted by four 3-D convolutional layers using four types of kernel sizes ($3 \times 3 \times 3$, $4 \times 4 \times 2$, $5 \times 5 \times 4$, and $6 \times 6 \times 5$) to obtain four types of feature maps with different scales. A 3-D convolutional layer with a kernel size of $2 \times 2 \times 1$ is added after the first and third 3-D convolutional layers, and therefore, the four feature matrices can be combined into spatial features. Then, we concatenate the extracted spatial features and spectral features as the input of the detection subnetwork.

The framework of the detection subnetwork is shown in Fig. 5(c), including a reshaping layer, a 2-D convolutional layer, a maximum pooling layer, three fully connected layers, and a softmax layer. The 3-D feature is reshaped into a 2-D feature matrix in the reshaping layer using a 3-D convolutional layer with a kernel size of $3 \times 3 \times 15$, which can fully utilize the spatial and spectral features extracted previously. The features are then extracted using a conventional 2-D feature extraction network; after that, the softmax layer is used to output the detection map.

The spectral–spatial detection network is trained using the cross-entropy loss function, and network weight parameters are optimized using the SGD. The input and output data shapes of each module in the network are shown in Table I.

IV. Analysis and Discussion of Results

In this section, we will demonstrate the effectiveness of the TDSS-UTD framework and its associated analysis. The experimental data, the experimental environment, and the evaluation metrics for this research are introduced at the very beginning. After that, we demonstrate the experimental effects of each network in the TDSS-UTD framework and compare the proposed method to the classical anomaly detection methods. Finally, the effects of parameter settings and the robustness of our model at various SNR interferences are analyzed and discussed.

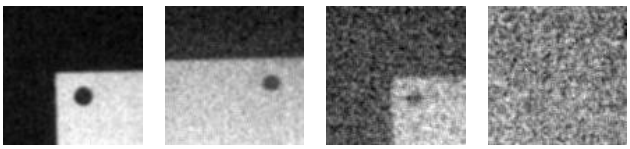
A. Data Description

1) Pool Dataset: The experimental data were obtained in the anechoic pool laboratory of the School of Navigation, Northwestern Polytechnical University, Xi'an, China. A Gaia Field portable spectral imager with a spectral resolution of 2.8 nm and a spectral band range of 400–1000 nm was utilized to capture the spectrum. We selected the 420–700-nm band

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TABLE I
INPUT AND OUTPUT DATA SHAPE OF EACH NETWORK LAYER IN THE SPECTRAL–SPATIAL DETECTION NETWORK (ILLUSTRATION WITH A NEIGHBORHOOD FACTOR OF 9)

Network layer	Input data shape	Output data shape
Spectral features extraction subnetwork		
2×2 × 2 Conv3D	(1024,1,9,9,120)	(1024,10,10,10,61)
2 × 2 × 2 Conv3D	(1024,10,10,10,61)	(1024,16,9,9,60)
2 × 2 MaxPooling	(1024,16,9,9,60)	(1024,16,8,8,59)
3 × 3 × 2 Conv3D	(1024,16,8,8,59)	(1024,16,9,9,60)
2 × 2 AvgPooling	(1024,16,9,9,60)	(1024,16,8,8,59)
3 × 3 × 2 Conv3D	(1024,16,8,8,59)	(1024,32,8,8,60)
Cat	(1024,32,8,8,60) (1024,32,8,8,60)	(1024,32,8,8,120)
Spatial features extraction subnetwork		
2×2 × 2 Conv3D	(1024,1,9,9,120)	(1024,32,8,8,41)
2 × 2 × 2 Conv3D	(1024,1,9,9,120)	(1024,32,9,9,40)
2 × 2 MaxPooling	(1024,32,9,9,40)	(1024,32,8,8,40)
3 × 3 × 2 Conv3D	(1024,1,9,9,120)	(1024,32,7,7,40)
2 × 2 AvgPooling	(1024,32,7,7,40)	(1024,32,8,8,42)
3 × 3 × 2 Conv3D	(1024,1,9,9,120)	(1024,32,8,8,40)
Cat	(1024,32,8,8,41) (1024,32,8,8,40 (1024,32,8,8,42) (1024,32,8,8,40)	(1024,32,8,8,163)
Detection subnetwork		
Cat	(1024,32,8,8,120) (1024,32,8,8,163)	(1024,32,8,8,283)
3×3 × 15 Conv3D	(1024,32,8,8,283)	(1024,40,1,1,54)
Reshape	(1024,40,1,1,54)	(1024,1,40,54)
3 × 3 Conv2D	(1024,1,40,54)	(1024,16,38,52)
2 × 2 MaxPooling	(1024,16,38,52)	(1024,16,19,26)
Flat	(1024,16,19,26)	(1024,7904)
Fc	(1024,7904)	(1024,2)
Sigmoid	(1024,2)	(1024,2)



(a) (b) (c) (d)

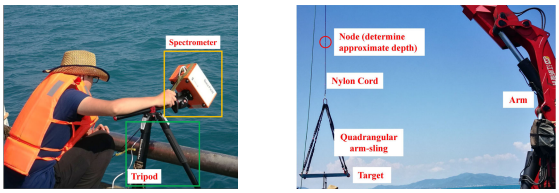
Fig. 6. Hyperspectral images in pool dataset at several depths (at the 683-nm band). (a) 0.1 m. (b) 0.5 m. (c) 1.0 m. (d) 1.6 m.

range in the analysis process, with a total of 120 wavebands chosen for the analysis. The light source used in the experiment was natural light. The hyperspectral data were calibrated by a standard whiteboard during the experimental process. The calibrated reflectance data were generated as follows:

$$r(x, y, \lambda) = \frac{\mathrm{DN}_{\mathrm{pixel}}(x, y, \lambda)}{\mathrm{DN}_{\mathrm{ref}}(x, y, \lambda)} \rho_{\mathrm{ref}} \tag{13}$$

where $\mathrm{DN}_{\mathrm{pixel}}(x, y, \lambda)$ and $\mathrm{DN}_{\mathrm{ref}}(x, y, \lambda)$, respectively, denote the pixel brightness value and the brightness value of the standard whiteboard, respectively, and ρ_{ref} represents the known reflectance value of the standard whiteboard. Here, we choose the 99% standard white plate. We only display four representative HSI in Fig. 6. We also divide the pool dataset into training and test sets in a ratio of 1:1.

2) Sea Dataset: The environmental factors under real seawater are more complicated, such as sea surface waves,



(a) (b)

Fig. 7. Experimental equipments for acquiring the sea dataset. (a) Spectrometer. (b) Experimental equipments.



(a) (b)

Fig. 8. False-color images of underwater iron plate at different depths in the sea dataset. (a) 0.8 m. (b) 3.0 m.

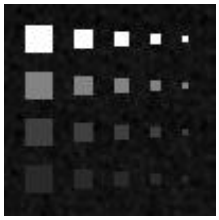


Fig. 9. Synthetic HSI (at the 683-nm band).

seawater quality, and target swaying. In order to prove the feasibility of our method in practical applications, we conducted experiments in Sanya, Hainan Province ($18°17'8.9''N, 109°5'10.6''E$). The iron plate targets were placed at the depths of 0.8 and 3.0 m in sea. The experimental equipment for acquiring the sea dataset is shown in Fig. 7, where we used a node on the nylon cord to determine the depth and a Gaia Field spectrometer to acquire HSIs of the underwater targets. Due to the influence of sea surface fluctuation and target swaying in the water, the acquired hyperspectral underwater target images are complex shapes, as shown in Fig. 8.

3) Synthetic Dataset: Following Section II, we used IOPE-Net to predict water properties absorption $a(\lambda)$ and scattering $b_b(\lambda)$. Given the depth range of 0–1.9 m likely to the real data, we employed the underwater RTE to construct a synthetic dataset, which is labeled with the corresponding depth value. As shown in Fig. 1, as the depth increases, the reflectance of the spectra gradually decreases. More information about how this dataset is utilized can be found in Section III.

4) Synthetic HSI: The synthetic HSI is 100 × 100 pixels in size and has 120 bands with a spectral band range of 400–740 nm. The area of this image that consists of pure water was obtained by directly taking a snapshot from the pure water region of the 1.9-m pool dataset. Target (we served iron as the target in this article) was then embedded at various depths and pixel sizes, including 0.1, 0.5, 1.0, and 1.6 m and 13 × 13,

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9×9, 7×7, 5×5, and 3×3. Further details of the synthetic HSI are shown in Fig. 9.

B. Experimental Setup

1) Comparison Methods: To further confirm the effectiveness of the TDSS-UTD framework, we compare the TDSS-UTD framework to five classical methods, including the three traditional methods ACE [49], RX [25], and MF [26], as well as the improved approaches, HSI-CNN [37] and modified spectral angle mapper (MSAM) [50].

2) Evaluation Metrics: Both qualitative and quantitative evaluations are employed so that the performance of several models can be compared. We select several evaluation metrics that are commonly employed in target detection, such as the receiver operating characteristic (ROC) curve and area under the ROC curve (AUC) value. These metrics, which are presented as follows, evaluate the accuracy and precision of the target detection model.

According to the output of our detection network, we raised the classification threshold from 0 to 1 to plot the ROC curve. Each time, we calculated the true positive rate (TPR) and the false positive rate (FPR), which were utilized as the horizontal and vertical coordinates to obtain the ROC curve. The TPR and FPR are defined as

$$\mathrm{TPR} = \frac{\mathrm{TP}}{\mathrm{TP} + \mathrm{FN}} \tag{14}$$

$$\mathrm{FPR} = \frac{\mathrm{FP}}{\mathrm{TN} + \mathrm{FP}}. \tag{15}$$

In addition, we utilize the AUC value to compare the detection performance more comprehensively. Assuming that the ROC curve is specified as a concatenation of points with coordinates of (x_1, y_1) and (x_2, y_2), the AUC can be determined by summing the areas of the components under the ROC curve. It can be calculated as

$$\mathrm{AUC} = \frac{1}{2}\sum_{i=1}^{m-1}(x_{i+1} - x_i)(y_i + y_{i+1}). \tag{16}$$

3) Experimental Environment: Our experiments are based on Intel[1] Core[2] i9-12900KF CPU, GeForce RTX 3080 Ti GPU, 64-GB RAM, and Windows 11 OS, and the related code is developed based on Python 3.9 and the deep learning framework PyTorch 1.12.1. Whiteboard correction has been utilized to preprocessing of the hyperspectral data. The batch size in batch training is 1024. There have been 500 training iterations. The rate of learning is 0.0001. The training algorithm is the SGD algorithm, and the hybrid loss functions are the evaluation criterion mentioned in Section III.

C. Results Analysis

1) Effectiveness Evaluation of the Data Transferring Network: In Section III, we proposed the data transferring network. Its function is to remove noise interference by transferring the real data to noise-free synthetic data. Therefore, for the sake of confirming the effectiveness of this specific network, we conduct the real spectral curves with transferred spectral curves as the comparison. As shown in Fig. 10, we choose one real spectrum, two synthetic spectra, and one spectrum that has been transferred by the data transferring network to test the effectiveness of the network at depths of 0.1, 0.5, and 1.6 m. There has been a significant divergence between the real and synthetic spectra in the 500–600-nm band range and both the amplitudes of the real and synthetic spectra decline noticeably with increasing depth. In principle, the synthetic spectra represent the theoretical reflectance value of iron at the current water properties and depth, whereas the real spectra are the actual hyperspectral data observed by the sensor. The real spectra are disturbed during instrumentation and underwater optical transmission, which leads to the distinction. This highlights the necessity of the proposed data transferring network, which transforms real spectra into synthetic spectra to lessen noise interference and improve target detection precision. In practical applications, it is recommended to exploit the data transferring network before target detection.

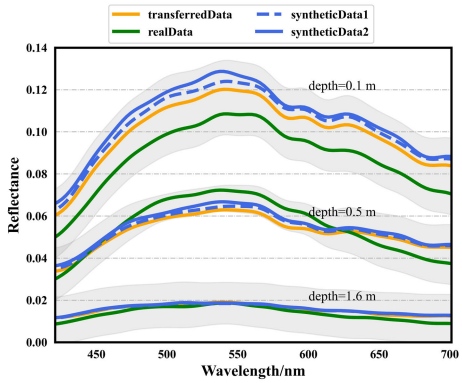


Fig. 10. Relationship between the real spectral curves, synthetic spectral curves, and transferred spectral curves at three depths. The transferred spectral curves are closer to the synthetic spectral curves. This demonstrates the effectiveness of the data transferring network in reducing noise.

TABLE II
CORRELATION ANALYSIS AND ERROR STATISTICAL ANALYSIS OF THE TRUE AND PREDICTED DEPTH VALUES

Range of Depth/(m)	0.1–1.0	1.3	1.6	1.9	Average
Correlation coefficient	0.994	0.726	0.727	0.615	0.766
Error within 0.05 m	77.0%	74.9%	55.2%	59.4%	66.4%
Error within 0.10 m	100%	98.9%	92.9%	92.9%	96.2%

2) Effectiveness Evaluation of the Depth Estimation Network: Considering that underwater target spectra highly depend on their depth in water, we came up with a depth estimation network to forecast the depth of underwater targets. The spectral–spatial detection network is then selected based on the predicted depth.

The performance of the depth estimation network is evaluated in the synthetic dataset. From the synthetic dataset, some test spectra are collected at depths of 0.1–1.0, 1.3, 1.6, and 1.9 m, and these data are labeled with the depth of the target. To evaluate the effectiveness of the depth estimation network, we statistically compare the true depth values to the predicted depth values.

We illustrate the real and predicted values in Fig. 11(a)–(d). The data points lie exactly on the orange line $y = x$ in the

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TABLE III
AUC VALUES COMPARISON FOR ALL EXPERIMENTAL DATA. THE BEST AUC VALUES IN EACH ROW ARE SHOWN IN BOLD

Dataset	RX	ACE	MF	HSI-CNN	MSAM	TDSS-UTD
0.1 m pool dataset	0.981	0.502	0.807	0.998	0.987	**1.000**
0.5 m pool dataset	0.951	0.492	0.753	0.998	0.982	**1.000**
0.8 m pool dataset	0.911	0.555	0.853	0.999	0.986	**1.000**
1.0 m pool dataset	0.827	0.515	0.819	0.995	**1.000**	0.999
1.3 m pool dataset	0.720	0.751	0.925	0.750	0.992	**1.000**
1.6 m pool dataset	0.592	0.590	0.874	0.583	0.991	**0.996**
Synthetic HSI	0.460	0.546	0.499	0.861	0.745	**0.962**
Average	0.777	0.568	0.839	0.883	0.990	**0.994**

TABLE IV
AUC VALUES COMPARISON FOR THE POOL DATASET WITH SNRS OF 12, 15, 17, AND 20. THE BEST AUC VALUES ARE SHOWN IN BOLD

Methods	0.8 m pool dataset						1.6 m pool dataset					
	RX	ACE	MF	HSI-CNN	MSAM	TDSS-UTD	RX	ACE	MF	HSI-CNN	MSAM	TDSS-UTD
$SNR = 12$	0.625	0.665	0.966	**0.987**	0.986	0.931	0.563	0.607	0.893	0.588	0.988	**0.991**
$SNR = 15$	0.687	0.621	0.940	**0.994**	0.986	0.826	0.582	0.609	0.890	0.592	0.989	**0.994**
$SNR = 17$	0.730	0.607	0.923	0.992	0.986	**0.998**	0.584	0.602	0.885	0.592	0.990	**0.995**
$SNR = 20$	0.806	0.588	0.896	0.996	0.986	**0.997**	0.595	0.595	0.881	0.589	0.990	**0.996**

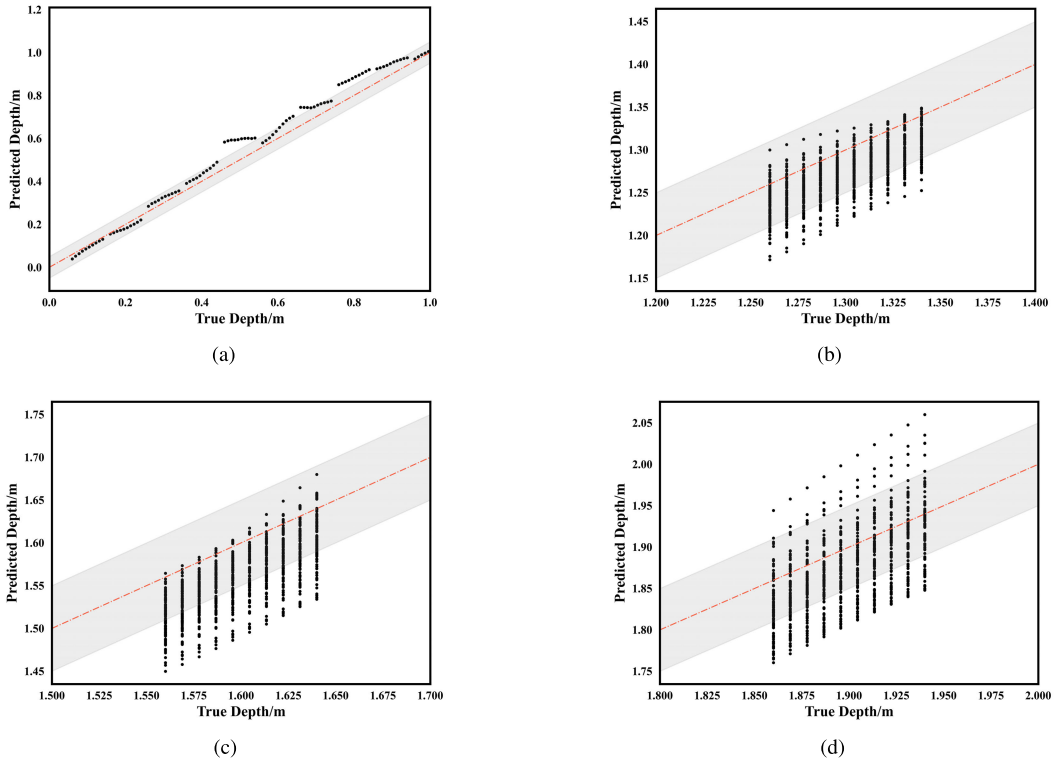


Fig. 11. True depth values versus predicted depth values at several depths. (a) 0.1–1.0 m. (b) 1.3 m. (c) 1.6 m. (d) 1.9 m. The points on the orange line represent that the predicted and true depths are equal. The data points in the gray-shaded area present that the predicted depth values have no impact on the subsequent processing, although they are not particularly accurate.

figure, which indicates that the predicted and true values are equal. The more closely the predicted value is aligned with the actual value, the closer the data points are to the orange line. The discrepancy between the predicted depth and the actual depth can be utilized to analyze the validity of the depth estimation network. The correlation analysis and the erroneous statistics analysis for the actual and predicted depth values are represented in Table II.

The correlation coefficient between the true depth value and the predicted depth value in the 0.1–1.0-m depth range is 0.994, which shows highly linear, and the error of 100% of the data is controlled within 0.10 m. Therefore, it will not influence the determination of the most appropriate spectral–spatial detection network.

The average correlation coefficient is 0.766 at all depths, with an essentially linear relationship. As shown in Fig. 11,

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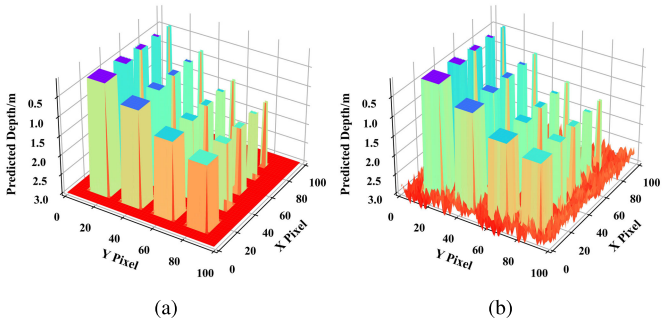


Fig. 12. Depths of the synthetic HSI. (a) Predicted depth graph. (b) True depth graph.

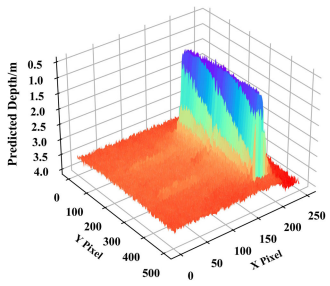


Fig. 13. Part of the contour information of the cylindrical iron block at 1.3 m acquired from the depth estimation network.

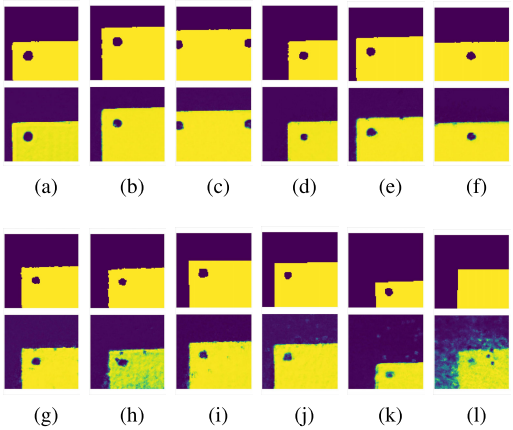


Fig. 14. Detection effect plots of the pool dataset with depths ranging from 0.1 to 1.6 m. The first row shows the ground truth, while the second row depicts the predicted probability map. (a) 0.1 m. (b) 0.2 m. (c) 0.3 m. (d) 0.4 m. (e) 0.5 m. (f) 0.6 m. (g) 0.7 m. (h) 0.8 m. (i) 0.9 m. (j) 1.0 m. (k) 1.3 m. (l) 1.6 m.

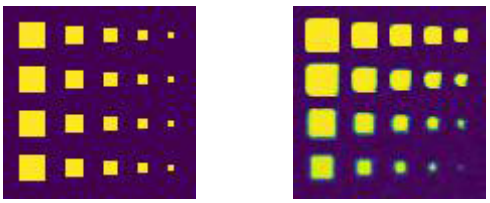


Fig. 15. Detection effect graph of synthetic hyperspectral data. (a) Ground truth. There are four rows of iron in the graph, and the depths at which they are located are 0.1, 0.5, 1.0, and 1.3 m, from top to bottom. To test the impact of the target size on the model's detection effect, there is just a size difference from left to right in each row, while the depth is maintained. (b) Predicted probability map. It can be observed that the prediction probability map clearly illustrates the phenomenon of "rendering misinterpretation."

the error on 77.1% of the data in the 0.1–1.0-m depth range is controlled within 0.05 m (falling in the shaded area), which can correctly choose the most appropriate spectral–spatial detection network. The average error on 94.9% of the data in the 1.3-, 1.6-, and 1.9-m depths is controlled within 0.10 m, which can also correctly choose the most appropriate spectral–spatial detection network. However, the remaining data will select a slightly deviated spectral–spatial detection network, which will make the target detection results not achieve the best results. As the training sets of the adjacent-depth spectral–spatial detection network only differ slightly in magnitude, the prediction performance of the adjacent-depth spectral–spatial detection network is not significantly different from the best network.

We predict the depth value of each pixel in the synthetic HSI to obtain the predicted depth map corresponding to the entire HSI, and we draw the real depth map and the predicted depth map as a 3-D graph representing the depth information, as shown in Fig. 12. This further demonstrates the effectiveness of the depth estimation network. In addition, we utilized the depth estimation network on a cylindrical iron block whose depth is 1.3 m submerged in water. We can gain a general contour of the target's approximate shape, which is plotted in Fig. 13. These phenomena show the practicability of employing the depth estimation network to select finer detection network and obtain general underwater target contour information.

3) Effectiveness Evaluation of the Spectral–Spatial Detection Network: Considering the special dependency of the observed spectrum of the underwater target, the systematic combination of the 3-D convolution layer and the 2-D convolution layer is proposed in the spectral–spatial detection network for extracting spectral–spatial information in HSI. We will demonstrate the target detection results and their associated discussions. We detect the synthetic HSI and HSIs at each depth in the pool dataset using the spectral–spatial detection network. To further validate the excellent performance of our model, the spectral–spatial detection network is compared to the classical methods ACE, RX, MF, HSI-CNN, and MSAM mentioned previously. The ROC curves and AUC values are used as quantitative evaluation metrics to assess the performance of the model.

The detection results for the pool dataset and the synthetic HSI are sequentially shown in Figs. 14 and 15. In these plots, the first row is the ground truth, which indicates the real category (i.e., water or iron) of each pixel. The predicted probability maps, listed in the second row, reveal the probability value that each pixel belongs to the iron. The closer the probability value is to 1, the more likely the pixel is categorized as iron. The detection effect decreases as the depth increases, according to the visualization results of these Figs. 14 and 15. At greater depths, the difference between the spectral curves of iron and water is minimal, which causes some misinterpretation of the two. Notably, we observe that a small portion of the water surrounding the

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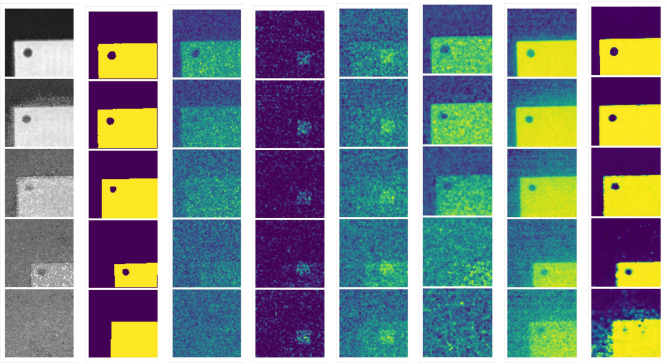


Fig. 16. UTD results of compared methods for the pool dataset at several depths. (a) Hyperspectral image. (b) Ground truth. (c) RX. (d) ACE. (e) MF. (f) HSI-CNN. (g) MSAM. (h) TDSS-UTD. The depths of the pool dataset are 0.1, 0.5, 1.0, 1.3, and 1.6 m, from top to bottom.

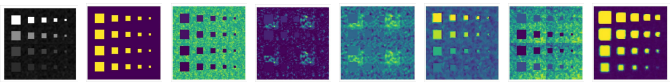


Fig. 17. UTD results of compared methods for the synthetic HSI. (a) Hyperspectral image. (b) Ground truth. (c) RX. (d) ACE. (e) MF. (f) HSI-CNN. (g) MSAM. (h) TDSS-UTD.

original iron is assimilated and recognized as iron, which is called "rendering misinterpretation." This is caused by the extensive neighborhood employed in this article. The cubic data with a huge neighborhood can also extract the spectral and spatial information of iron while detecting the edge pixels, leading to misinterpretation. However, the percentage of this rendering misinterpretation is negligible, and it is acceptable and tolerable in real-world scenarios.

To be specific, for the pool dataset at depths of 0.1, 0.5, 1.0, 1.3, and 1.6 m, we compare the detection results of the proposed spectral–spatial detection network with the conventional approaches, as shown in Fig. 16. The comparison on the synthetic HSI is also shown in Fig. 17. The MSAM approach predicts that the probability values corresponding to the pixels are mainly spread between 0.4 and 0.7, which cannot successfully discriminate between iron and water. The HSI-CNN method performed well between 0.1 and 1.0 m but did not detect well later. The RX anomaly detection method incorrectly detects water as the target and iron as the background. Our proposed method is superior to the classical methods when compared at all depths.

We also apply the framework proposed in this article to detect the target in the sea dataset. Water and the underwater targets have a shape of 350 × 350 after cropping. The experimental results are shown in Fig. 18. In the complex sea environment, it can be visualized that the detection of our proposed method TDSS-UTD is significantly better than those of other traditional or advanced hyperspectral detection methods. Light water waves are still detected by our method as water background instead of being misidentified as a target. The MSAM method can also faintly reveal that the target area, the HSI-CNN, MF, and ACE methods are worse, while the RX method is almost invalid. Therefore, this demonstrates how

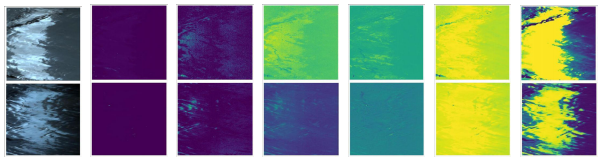


Fig. 18. UTD results of compared methods for the sea dataset at several depths. (a) Hyperspectral image. (b) RX. (c) ACE. (d) MF. (e) HSI-CNN. (f) MSAM. (g) TDSS-UTD. The depths of the sea dataset are 0.8 and 3.0 m, from top to bottom.

critical the spatial and spectral features that were extracted by our method are to the detection performance. Our method can still detect underwater targets in complex marine environments, which also proves the feasibility of our method in practical applications.

To quantitatively demonstrate the effectiveness of the detection performance, ROC curves and AUC values for the detection result were utilized as quantitative metrics. For pool datasets at depths of 0.1, 0.5, 0.8, 1.0, 1.3, and 1.6 m, the ROC curves of the detection effectiveness of the proposed spectral–spatial detection network and the classical methods are shown in Fig. 19. Moreover, Fig. 20 shows the ROC curves for the proposed spectral–spatial detection network and the conventional approaches for synthetic HSI. For the ROC curve, the closer the curve is to the top-left corner, the better the detection performance of the model is. It can be observed from Fig. 20 that the ROC curve of our proposed model is the closest to the top-left corner, indicating the best performance, both on the pool dataset and the synthetic HSI.

In addition, we compare the detection performance between our proposed approach and the classical methods in Table III using the quantitative metric AUC value. The higher the AUC value is, the better the detection performance of the model is. We can observe from Table III that among all the experimental data, our proposed TDSS-UTD framework has the highest AUC value. According to the above analyses, it is easy to find that our proposed framework has a great impact on the final UTD results.

D. Discussion

1) Parameter Sensitivity: The spectral feature extraction subnetwork and the spatial feature extraction subnetwork are the main part of the spectral–spatial detection network in our proposed framework. Thus, the neighborhood factor and the depth setting are vital parameters for target detection. To investigate the effect of these parameters on the effect of our model, we fix one parameter and dynamically adjust the other parameter. We exhibit the detection effect in Figs. 21–24. Figs. 21 and 23 show the effect of the neighborhood factor on the detection results for the pool dataset and synthetic HSI. The effect of the neighborhood factor on the AUC value in the pool dataset and synthetic HSI is shown in Fig. 23(a) and (b), respectively. It can be seen that the neighborhood factor has a few effects on the detection results in the pool dataset before 1.3 m, where the detection model is not sensitive to the neighborhood factor. To be specific, when the depth exceeds 1.3 m, the detection effect changes with the neigh-

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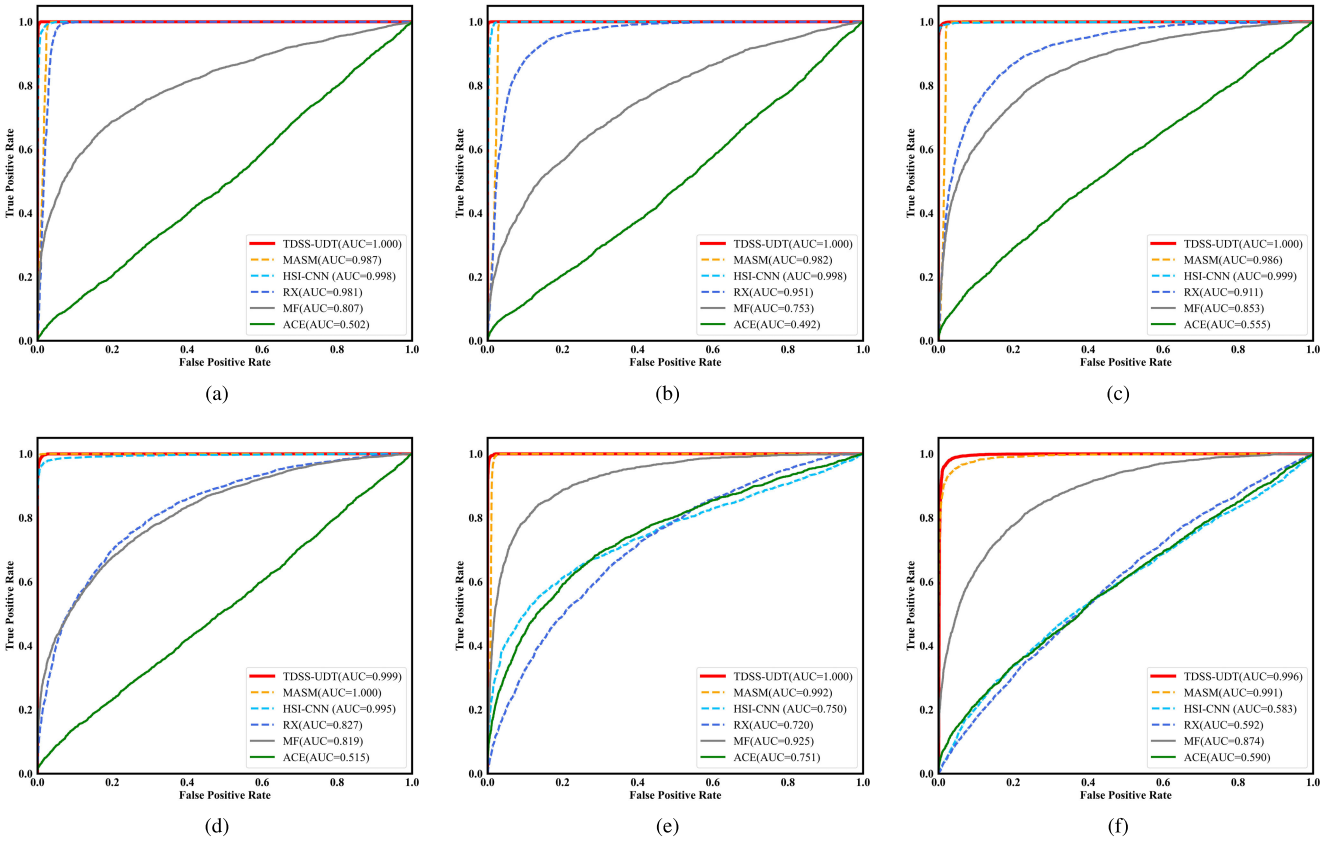


Fig. 19. ROC curves of the compared methods for the pool dataset. Depths of the pool dataset are (a) 0.1, (b) 0.5, (c) 0.8, (d) 1.0, (e) 1.3, and (f) 1.6 m. The curves of our proposed model are all the closest to the top-left corner, representing the best detection performance.

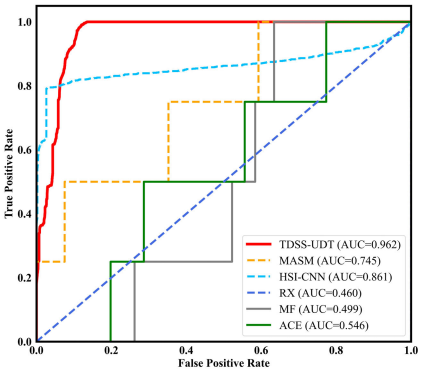


Fig. 20. ROC curves of the compared methods for synthetic HSI.

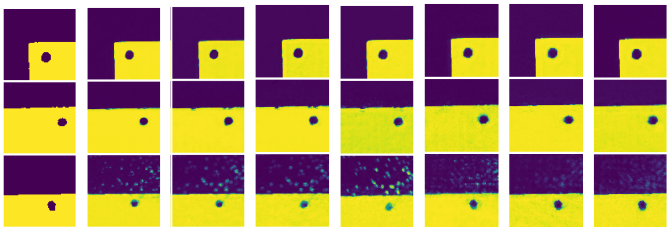


Fig. 21. Detection maps of the pool dataset with different depths and various neighborhood values. (a) Ground truth. Values of the neighborhood are (b) 5, (c) 7, (d) 9, (e) 11, (f) 13, (g) 15, and (h) 17. Three depth groupings, 0.1, 0.5, and 1.6 m, were established from top to bottom.

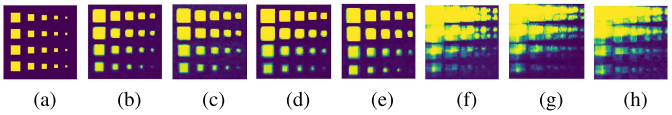
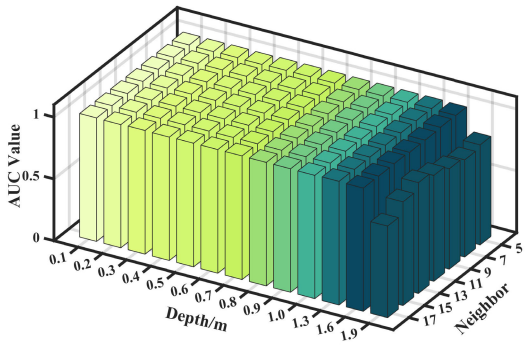


Fig. 22. Detection maps of the synthetic HSI with different neighborhood values. (a) Ground truth. Values of the neighborhood are (b) 5, (c) 7, (d) 9, (e) 11, (f) 13, (g) 15, and (h) 17.

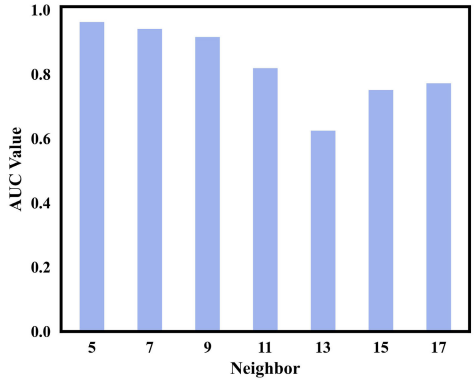
borhood factor. The results demonstrate that the detection effectiveness reaches its best when the neighborhood is 13. On the synthetic HSI, the highest AUC value is achieved when the neighborhood is 5. The lower the neighborhood value, the less "rendering misinterpretation" there is, and the better the detection performance since the targets embedded in the synthetic HSI are relatively compact.

To analyze the effect of the depth setting of the spectral–spatial detection network on the detection performance of our model, we fixed the neighborhood value to 5 and divided the depth set into two groups: the first group is [0.1, 0.2, 0.3, 0.4, 0.5, 0.6, 0.7, 0.8, 0.9, 1.0, 1.3, 1.6, 1.9] and the second group is a whole depth, where the network is trained for data at all depths. In Fig. 24, a comparison of the detection performance of the two groups is illustrated. The first group performs significantly better than the second group, which further demonstrates the necessity of the proposed depth estimation network and the effectiveness of the spectral–spatial detection network.

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(a)



(b)

Fig. 23. Sensitivity analysis of parametric neighborhood values. (a) Pool dataset. (b) Synthetic HSI.

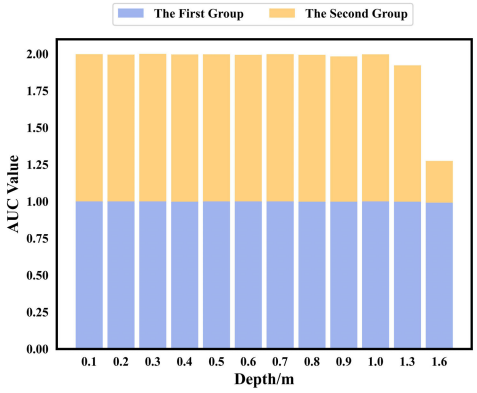


Fig. 24. Comparison of the AUC values of the two groups of spectral–spatial detection network.

2) Robustness Analysis: To further verify the robustness of our proposed framework, we employ the framework on underwater HSIs with various noise environments. We set the HSIs interfered with by noise at some given SNRs of 12, 15, 17, and 20. We use the quantitative metric AUC value to evaluate the detection effectiveness of the proposed framework and several classical methods in Table IV. The relationship between the AUC values and the SNR for the pool dataset at several depths is shown in Fig. 25.

It is effortless to find that, as the SNR exceeds 20, the detection performance of our model is substantially consistent with that in the absence of noise-free interference. Compared

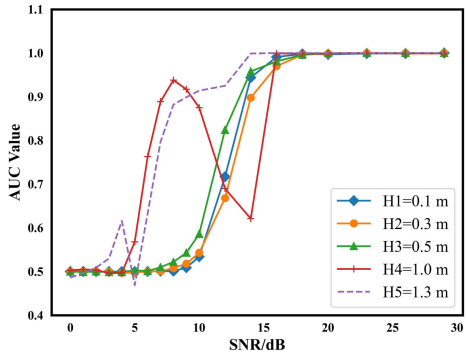


Fig. 25. Relationship between AUC values and SNR for the pool dataset at several depths.

with these classical methods, our proposed framework has more remarkable properties for removing noise interference, illustrating that our proposed model has excellent robustness, which makes our model more advantageous in UTD.

V. Conclusion

For detecting the underwater target from HSIs, we propose the TDSS-UTD framework in this article. It mainly consists of data transferring network, depth estimation network, and spectral–spatial detection network.

At the very beginning, we estimate water properties, which are utilized to construct the synthetic dataset and synthetic HSI for training our model. After that, the real hyperspectral data are transferred into the corresponding synthetic data by the data transferring network to eliminate noise interference. Moreover, the trained depth estimation network is also developed to predict the depth value of the spectra in water, which facilitates the selection of a more suitable spectral–spatial detection network. Finally, we segment the input HSI into several cubic data, extract the spectral–spatial features by the spectral–spatial detection network, and then obtain the detection map corresponding to the HSI. Moreover, the experimental results demonstrate that our proposed method can effectively detect underwater targets from HSIs. The TDSS-UTD framework can also attain incredibly high detection accuracy. In addition, it has outstanding robustness, which is more advantageous in UTD.

In terms of future development, it lies in: 1) the run time of the model will be cut in half and the real-time performance will be enhanced; 2) the overall model will be reconfigured as an end-to-end model for future applications; 3) a database will be constructed to provide water properties and target information in many water regions; and 4) more practical real data will be developed for experiment and the underwater noise distribution will also be developed. We believe that such a database can further contribute to making our model more applicable and robust.

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